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The Warping of Space and Time

in which Hermann Minkowski unifies space and time, and Einstein warps them

Minkowski's Absolute Spacetime

> The views of space and time which I wish to lay before you have sprung from the soil of experimental physics, and therein lies their strength. They are radical. Henceforth, space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality.

With these words Hermann Minkowski revealed to the world, in September 1908, a new discovery about the nature of space and time.

Einstein had shown that space and time are "relative." The length of an object and the flow of time are different when viewed from different reference frames. My time differs from yours if I move relative to you, and my space differs from yours. My time is a mixture of your time and your space; my space is a mixture of your space and your time.

Minkowski, building on Einstein's work, had now discovered that the Universe is made of a four-dimensional "spacetime" fabric that is absolute, not relative. This four-dimensional fabric is the same. as seen from all reference frames (if only one can learn how to "see" it); it exists independently of reference frames.

The following tale (adapted from Taylor and Wheeler, 1992) illustrates the idea underlying Minkowski's discovery.